

CHANDANA R

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Summary

Aspiring AI Engineer and Data Scientist with hands-on experience in Machine Learning, Deep Learning, Agentic AI, Retrieval-Augmented Generation (RAG), and Voice AI systems. Skilled in developing multi-agent AI applications, deep learning-based medical image classification models, predictive analytics solutions, and scalable backend services using Python, FastAPI, LangGraph, TensorFlow, and SQL. Passionate about building intelligent AI systems that solve real-world problems across healthcare, logistics, and enterprise automation.

Skills

Programming Languages: Python, SQL

AI Engineering: LangGraph, Multi-Agent Systems, RAG, ChromaDB, Prompt Engineering, FastAPI, REST APIs, WebSockets

Voice AI: Deepgram (Speech-to-Text), ElevenLabs (Text-to-Speech), Real-Time Streaming

Machine Learning: Regression, Classification, Feature Engineering, Data Preprocessing, Model Evaluation, Statistical Analysis

Deep Learning: CNN, Transfer Learning, TensorFlow, Keras, OpenCV, Grad-CAM

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, LangChain

Data Analytics: Excel, Power BI, Power Query, DAX, Dashboard Development, Exploratory Data Analysis (EDA)

Tools: Git, GitHub, Docker, Streamlit, VS Code, Jupyter Notebook

Core Concepts: Machine Learning, Deep Learning, Generative AI, NLP, System Design, Data Cleaning

Projects

AI – Agentic Voice AI for Logistics | *Python, FastAPI, LangGraph, ChromaDB, Deepgram, ElevenLabs, WebSockets* Jun 2024 – Present

- Built an enterprise-grade multi-agent Voice AI platform for logistics using LangGraph, enabling intelligent orchestration between Pricing, Distance, and Knowledge agents.
- Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and local embedding models, enabling accurate and context-aware responses from logistics policies and company documentation.
- Developed a custom logistics intelligence engine incorporating Haversine-based distance calculation, warehouse assignment, and GST-compliant pricing for automated shipment estimation.
- Designed FastAPI REST APIs and WebSocket streaming services to support real-time conversational AI with token-by-token response streaming.
- Integrated Deepgram Speech-to-Text and ElevenLabs Text-to-Speech to deliver an end-to-end voice AI assistant for logistics operations.
- Implemented conversation memory, structured logging using Loguru, and automated testing to improve system reliability, maintainability, and observability.

Brain Tumor Detection using Deep Learning (CNN + Grad-CAM) | *Python, TensorFlow, Keras, OpenCV, NumPy, Matplotlib* Mar 2024 – May 2024

- Developed an end-to-end deep learning system to classify brain MRI scans into Glioma, Meningioma, Pituitary, and No Tumor categories.
- Implemented transfer learning using VGG16, ResNet50, and MobileNetV2 to compare model performance and improve classification accuracy.
- Optimized training using staged transfer learning and selective fine-tuning, reducing overfitting and improving model convergence.
- Achieved a best test accuracy of **91.6%** using the VGG16 architecture with balanced classification across all tumor classes.
- Integrated Grad-CAM visualization to improve model interpretability by highlighting tumor regions influencing model predictions.
- Applied image preprocessing, feature learning, data augmentation, and performance evaluation techniques to develop a robust medical image classification pipeline.

Education

Indian Institute of Information Technology Dharwad (IIIT Dharwad)

Bachelor of Technology in Data Science and Artificial Intelligence

Dharwad, Karnataka

2021 – 2025

Achievements

- Built and deployed an end-to-end Multi-Agent Voice AI platform integrating LangGraph, FastAPI, Deepgram, ElevenLabs, and ChromaDB.
- Developed a CNN-based Brain Tumor Detection system achieving **91.6%** classification accuracy using VGG16 and Grad-CAM visualization.